CHAPTER 4

PRINCIPLES OF PROGRAMMING AND PROBLEM SOLVING

CHAPTER ESSENCE AT A GLANCE:
Master the formal methodology of software engineering from problem decomposition and algorithmic design to flowchart modeling, structured debugging, and comprehensive documentation.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Systematic Problem Solving Lifecycle:** Robust software engineering follows rigorous phases: problem definition, analysis, algorithmic design, coding, debugging, testing, and documentation.

• **Algorithmic Finiteness and Determinism:** An algorithm must be unambiguous, accept zero or more inputs, produce defined outputs, and terminate after a finite sequence of discrete steps.

• **Visual Flowchart Logic:** Standard geometric flowchart symbols represent distinct actions: rectangles for processing, parallelograms for I-O, and rhombuses for conditional decisions.

• **Error Classification Taxonomy:** Syntax errors violate language grammar at compile time, runtime errors crash execution dynamically, while logical errors yield incorrect results silently.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Development Lifecycle} = \text{Analysis} + \text{Algorithm} + \text{Coding} + \text{Testing}$$
Systematic Problem Solving Pipeline

$$\text{Algorithm Requirements} = \text{Input} + \text{Output} + \text{Definiteness} + \text{Finiteness}$$
Donald Knuth Algorithmic Criteria

$$\text{Decisions} \in \{\text{Rhombus: Branching}\} \quad | \quad \text{Rectangles: Processing}$$
Standard Flowchart Geometric Notation

$$\text{Error Types} \in \{\text{Syntax (Compile)}, \text{Runtime (Crash)}, \text{Logical (Wrong Output)}\}$$
Software Defect Classification Taxonomy

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In flowchart diagrams, decision diamond (rhombus) boxes MUST have at least two clearly labeled exit paths ('Yes'/'No' or 'True'/'False').

• **Alert:** Do not confuse logical errors (program runs to completion but gives incorrect output) with runtime errors (program crashes midway, e.g. division by zero).

• **Alert:** Input and output operations must always be enclosed in parallelograms, never in rectangular process boxes.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define an Algorithm. State four essential characteristics of an algorithm. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A finite step-by-step sequence of well-defined instructions designed to solve a specific computational problem. **[1 Mark]**

• **Characteristics:** (1) Input: accepts zero or more inputs; (2) Output: produces at least one result; (3) Definiteness: each step is clear and unambiguous; (4) Finiteness: terminates after a finite number of steps. **[1 Mark]**

2. Distinguish between Syntax Errors and Logical Errors with an example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Syntax Error:** Violation of the grammatical rules of the programming language detected by the compiler at compile time (e.g. omitting a semicolon, misspelling keywords). **[1 Mark]**

• **Logical Error:** Flaw in the programmer's reasoning where program compiles and runs successfully but generates incorrect output (e.g. using average = a + b divided by 2 instead of (a + b) divided by 2). **[1 Mark]**

3. Explain the major phases involved in systematic problem solving using computers. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Problem Definition & Analysis:** Clearly understand the problem boundary, identify required inputs, expected outputs, and constraints. **[1 Mark]**

• **Algorithm Design & Flowcharting:** Formulate step-by-step logic and depict data flow visually using standardized flowchart symbols. **[1 Mark]**

• **Coding, Debugging & Documentation:** Write code in high-level language, eliminate syntax and logical errors through testing, and create internal and external documentation. **[1 Mark]**

4. Draw standard flowchart symbols and state the specific purpose of each: Oval, Parallelogram, Rectangle, Rhombus, and Circle. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Terminal (Oval) & Process (Rectangle):** Oval marks Start and Stop points; Rectangle represents arithmetic calculations and internal data assignments. **[1 Mark]**

• **Input-Output (Parallelogram) & Decision (Rhombus):** Parallelogram handles reading input data and displaying output results; Rhombus evaluates conditional tests branching into two paths. **[1 Mark]**

• **Connector (Circle):** Connects different flow paths on the same page, preventing intersecting line clutter. **[1 Mark]**

5. Write a step-by-step algorithm to find the largest among three numbers A, B, and C. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 & 2:** Start program. Read three numbers A, B, and C. **[1 Mark]**

• **Step 3 & 4:** If A > B and A > C, then write 'A is largest'; Else if B > C, then write 'B is largest'; Else write 'C is largest'. **[1.5 Marks]**

• **Step 5:** Stop program execution. **[0.5 Mark]**

6. Develop an algorithm and write pseudocode to calculate the factorial of a given positive integer N. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Initialization):** Start. Read integer N. Initialize fact = 1 and counter i = 1. **[1 Mark]**

• **Step 2 (Looping Construct):** Repeat steps while i <= N: fact = fact * i; i = i + 1. **[1.5 Marks]**

• **Step 3 (Output & Boxed Result):** Display calculated fact value. Stop. Boxed Result: fact = N! accumulated iteratively. **[1.5 Marks]**

7. Explain Debugging techniques and contrast Runtime Errors with Compile-time Errors with examples.
[Score: 3 Marks]

Answer & Key Valuation Steps:

• **Debugging Definition:** The systematic process of detecting, isolating, and fixing defects or bugs that prevent software from running correctly. **[1 Mark]**

• **Compile-time Errors:** Detected by the compiler prior to executable generation (e.g. type mismatch, missing declarations, syntax errors). **[1 Mark]**

• **Runtime Errors:** Errors occurring during actual program execution causing unexpected program crashes (e.g. dividing by zero, accessing null pointers, memory allocation failure). **[1 Mark]**